

ECE 647: Lecture 8

1 Operations that Define Convexity

1.1 non-negative weighted sums

Let $f_1, f_2, \dots, f_m$ be convex, and $w_1, w_2, \dots, w_m \geq 0$.

$\implies w_1 f_1 + w_2 f_2 + \dots + w_m f_m$ is also **convex**.

Proof. By definition. □

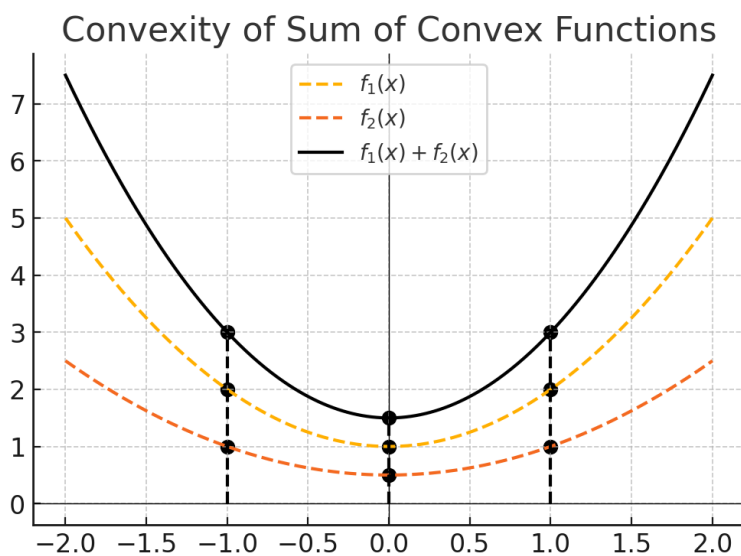


Figure 1: Convexity of weighted sum of convex functions

Can be extended to infinite sums and integrals. $f(x, y)$ is convex in x for each y and $w(y) \geq 0$. Then

$$g(x) = \int_A w(y) f(x, y) dy$$

is **convex** in x . Similarly, non-negative weighted sums of **concave** functions is **concave**.

Q: What if some of the weights are negative?

No, the sum of convex functions is **not necessarily convex** if some of the weights are negative.

1.2 Affine mapping of the argument

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $\mathbf{A} \in \mathbb{R}^{n \times m}$, and $\mathbf{b} \in \mathbb{R}^n$.

Define: $g : \mathbb{R}^m \rightarrow \mathbb{R}$ by

$$g(\mathbf{x}) = f(\mathbf{Ax} + \mathbf{b})$$

with $\text{dom}(g) = \{\mathbf{x} | \mathbf{Ax} + \mathbf{b} \in \text{dom}(f)\}$.

Then if

- if f is convex, so is g ,
- if f is concave, so is g .

Proof. By definition. □

Example 1. $\log(\sum_{i=1}^n e^{x_i})$ is convex, then $\log(e^{x_1+2x_2} + e^{2x_2+x_3} + e^{3x_1+x_3})$ is convex.

Example 2. x^2 is convex, then $2(x_1 + 3x_3)^2 + (x_2 + x_4)^2$ is convex.

Q: If $f(x)$ is convex, is $f(-x)$ convex? Yes.

Q: If $f(x)$ is convex, is $-f(x)$ convex? No.

Using the above two properties, whenever we check for convexity, we can ignore the affine changes of variables and non-negative weights, and focus on the simplest function form.

To use convex optimization, it is very important to be able to quickly identify/convert to convex functions. A bit exaggerating, but for some problems, the key to success is to find (identify) convex functions! In the next section, we will provide some commonly-used convex functions with applications.

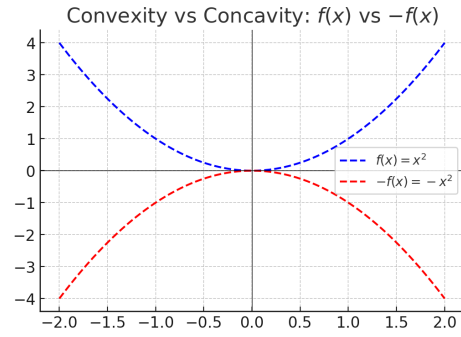
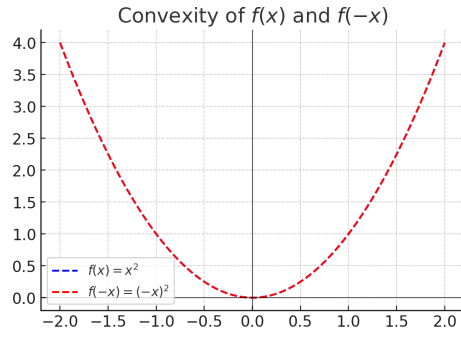


Figure 2: Convexity of $f(x)$ and $f(-x)$. **Figure 3:** Convexity of $f(x)$ and $-f(x)$.

2 Commonly used Convex Functions

2.1 Exponent

$$\begin{aligned} f(x) &= e^{ax}, \\ \frac{df(x)}{dx} &= ae^{ax}, \\ \frac{d^2f(x)}{dx^2} &= a^2e^{ax} \geq 0. \end{aligned}$$

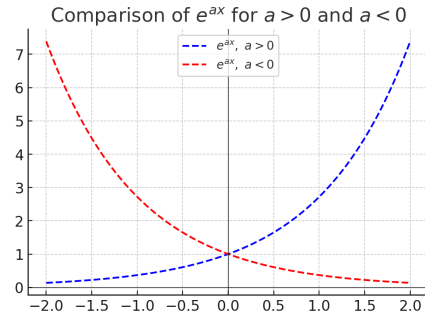


Figure 4: Exponential function.

2.2 Negative Logarithm

$$\begin{aligned} f(x) &= -\log(x), \\ \frac{df(x)}{dx} &= -\frac{1}{x}, \\ \frac{d^2 f(x)}{dx^2} &= \frac{1}{x^2} \geq 0. \end{aligned}$$

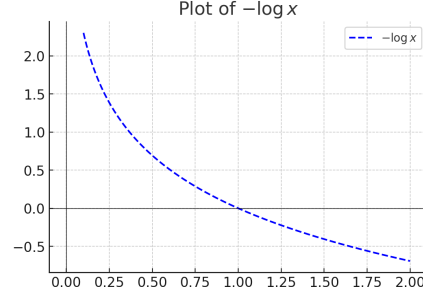


Figure 5: Negative logarithm function.

Example 3. *Shannon capacity*

$$C = W \log \left(1 + \frac{P}{N} \right)$$

- C is a **concave** function of P .
- Capacity increment due to increased power **diminishes** as $P \uparrow$.
- Diminishing marginal returns of C in P .

Example 4.

$$P = N \left(e^{\frac{C}{W}} - 1 \right)$$

- P is a **convex** function of C .
- To get the same increment in capacity, the required increase in power grows **quickly** as $C \uparrow$.

2.3 Power

$$f(x) = \begin{cases} x^a, & a \geq 1 \\ -x^a, & 0 \leq a \leq 1 \\ x^a, & a < 0 \end{cases}$$

Restricted domain to $(0, +\infty)$ and check second order derivative.

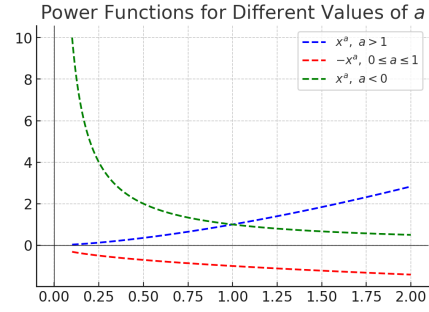


Figure 6: Power functions.

Power of Absolute Values

$$f(x) = |x|^p, \quad p \geq 1.$$

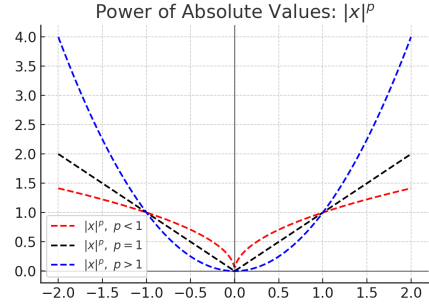


Figure 7: Power of absolute value.

Norms

$$f(x) = \|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{\frac{1}{p}}, \quad p \geq 1$$

Parabola

$$f(\mathbf{x}) = (\mathbf{x} - \mathbf{x})^T \mathbf{P} (\mathbf{x} - \mathbf{x}),$$

where $\mathbf{P}$ is positive semi-definite.

Example 5. Mean square errors and regression.

Given (x_i, y_i) for $i = 1, 2, \dots, N$, what are the parameters a, b such that the linear model $y = ax + b$ has the smallest mean square error, given by

$$\sum_{i=1}^N [y_i - (ax_i + b)]^2.$$

Q: Can we use other types of norm?

A: Yes, e.g.

$$\sum_{i=1}^N |y_i - (ax_i + b)|^p$$

This formulation is still convex when $p \geq 1$.

Example 6. Detection and likelihood functions

Take x as a parameter. Given x , the distribution of y is given by $p_X(y)$.

Problem: Estimate the value of parameter x based on observing one sample y from the distribution.

e.g. x is the transmitted signal and y is the received signal corrupted by Gaussian noise.

$$y = x + z,$$

$$p(z) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{z^2}{2\sigma^2}}.$$

Now given y , what is the estimate of x ?

Let $l(x) = \log(p(y|x))$ be the log-likelihood function. **Maximum likelihood detection**

Estimate x as

$$\hat{x} = \arg \max_x p(y|x) = \arg \max_x l(x)$$

i.e., , to choose as our estimate a value of the parameter that maximizes the likelihood function of the observed value of y .

For Gaussian noise,

$$p(y|x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-x)^2}{2\sigma^2}},$$

$$\Rightarrow l(x) = \log \frac{1}{\sqrt{2\pi\sigma^2}} - \frac{1}{2\sigma^2} (y-x)^2,$$

which is a **concave** function of x . Other forms of noise are also possible.

Example 7. Cost of energy generation

$$C(p) = ap + bp^2$$

Example 8. Utility function of an electricity consumer

$$U(p) = U_0 - b(p - p_0)^2$$

where p_0 is the most desirable consumption.